

BLE scan · beacons

2.4 GHz

Wi-Fi scan · APs

2.4 GHz

Wi-Fi scan · APs

5 GHz

Classic BT + BLE L2CAP

two-node counter link

Per-link DSP — production Kotlin classes

RingWindow · Stats (median, robust dispersion) · BaselineTracker · DipDetector / PeakDetector

noise-aware floor · sustained gate

Feature extraction

depth · duration · texture
variance · prominence · coherence

dBm → metres

log distance per band
gate -75 dBm · $f \in [0.15, 0.92]$

Scan-epoch delta

store → reset → rescan
stable level = median of last 20

EntityClassifier v1.1 → entity · confidence · event record

Counters · range band · field verdict · transit count · mesh confirmation · XP

artefacts: tables + logs